

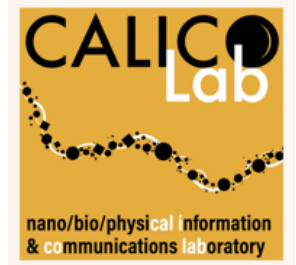
Physical and Chemical Reservoir Computing with Microfluidics and Molecular Communication

ACM NanoCom 2026 Tutorial: Artificial Intelligence in Microfluidic Circuits

22 September 2026

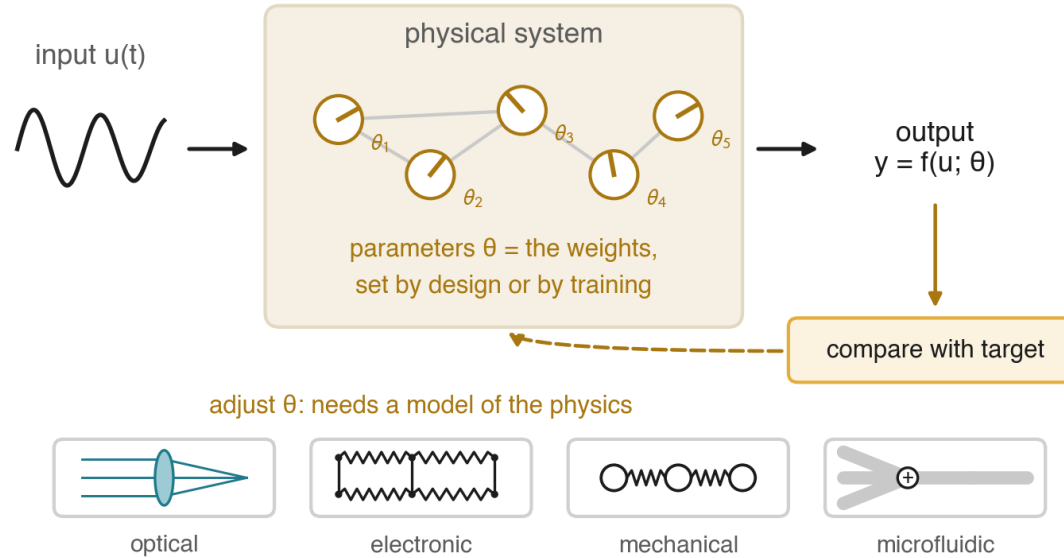
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Two ways to compute with physics

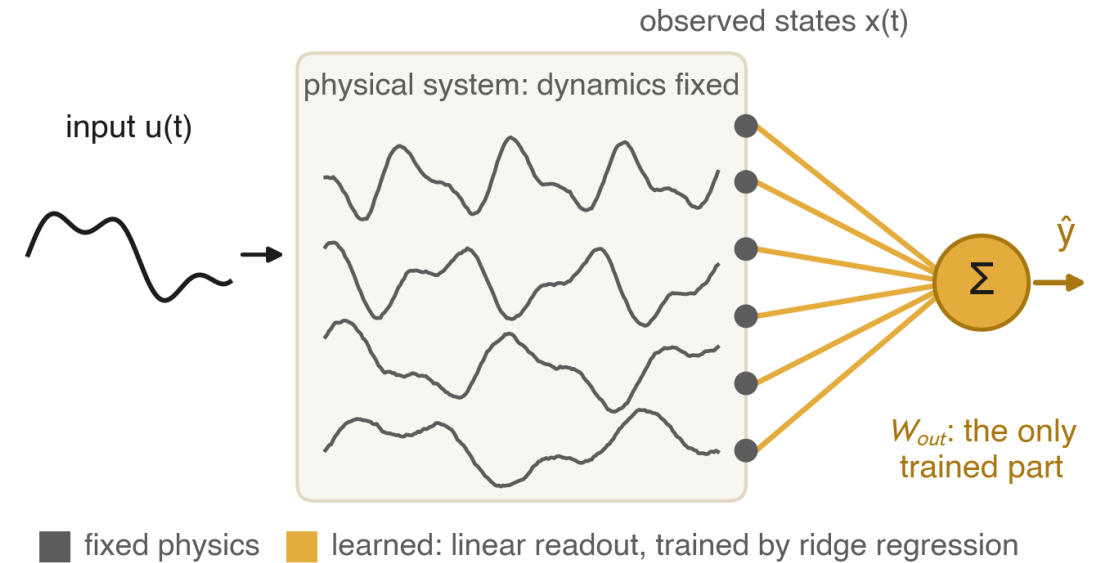
Train the physics *bottom-up: the physics is the network*



- **Weights are physical parameters**, designed or trained.
- **Training needs a model of the physics**, or physics-aware training.
- **Any architecture**, but retuning means changing the hardware.

What carries the program: the physical parameters. To change it, rebuild.

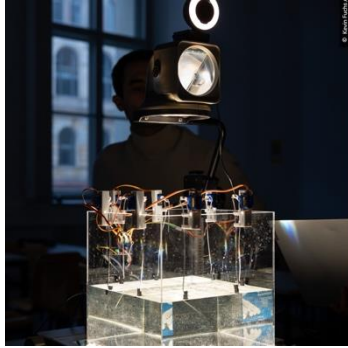
Physical reservoir computing *top-down: the system is a black box*



- **Nothing inside is designed or trained:** the dynamics are whatever the physics gives.
- **No model of the inside is needed**, only observations of its state.
- **The readout is a linear map W_{out}** , trained by ridge regression; retune by changing physical parameters.

What carries the program: the readout weights. To change it, retrain.

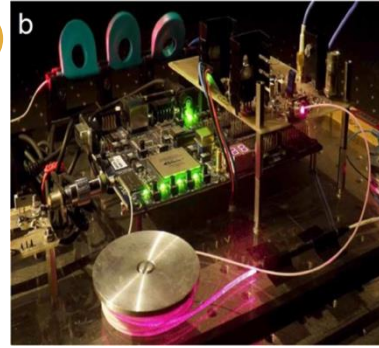
1



A bucket of water

Motors tap the surface; the wave pattern is the state, read by a camera and a linear classifier. Fernando & Sojakka 2003

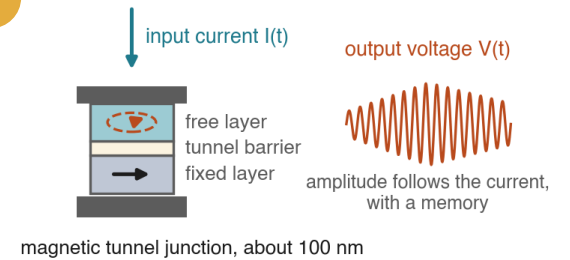
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Light in a fibre loop

Light circulates in a delayed feedback loop; one photodiode, sampled in time, is the state. Brunner et al. 2013

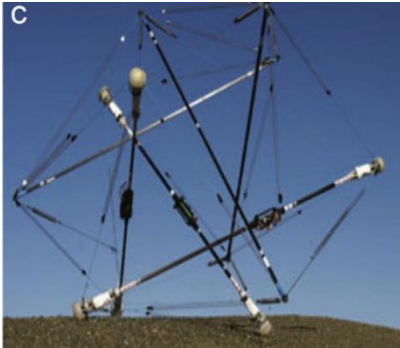
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A spintronic oscillator

A current drives a magnetic nano-oscillator; its oscillation amplitude in time is the state. Torrejon et al., Nature 2017

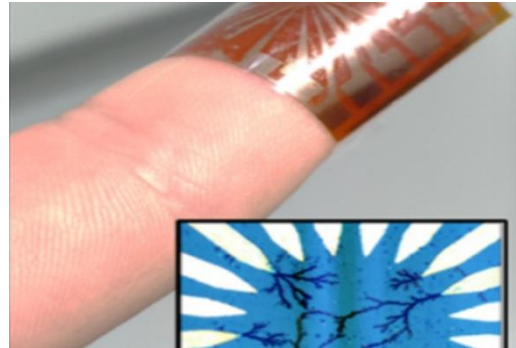
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A soft robot's body

Motor commands deform a tensegrity body; tension sensors give the state to a linear controller. Caluwaerts et al. 2014

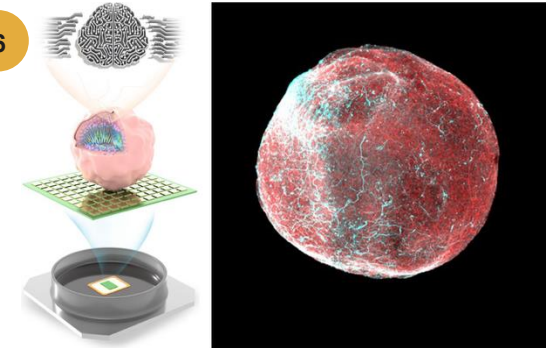
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A network of organic transistors

Voltages drive a network grown on plastic; currents from several terminals are the state. Cucchi et al., Sci. Adv. 2021

6

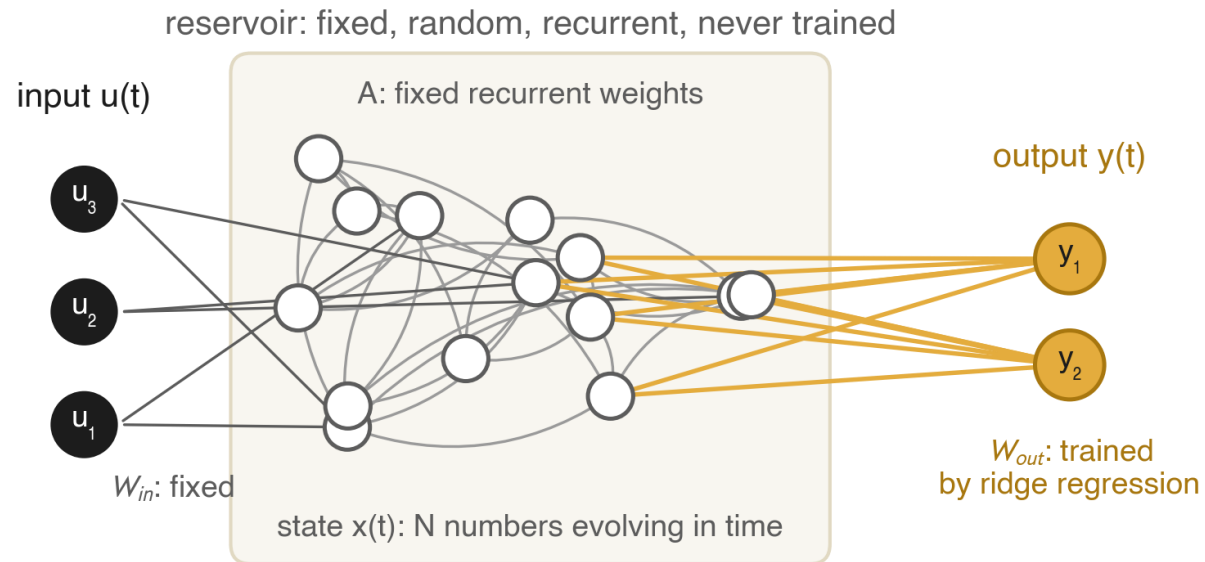


A brain organoid

A living neural network on an electrode array; pulses go in, the recorded spikes are the state. Cai et al., Nat. Electron. 2023

Any physical system with memory and nonlinearity can be a reservoir: the physics stays fixed, only the readout is trained.

Reservoir computing in one picture



- **Fixed and random:** A (recurrent) and W_{in} (input). Trained: only W_{out} .
- **Training is ridge regression:** regularised linear least squares with a closed-form solution; no back-propagation through time.
- **The reservoir is never modified**, so a physical system can be the reservoir.
- **One reservoir, many tasks:** the same state vector can feed several readouts, one per task.
- **Why it works:** the reservoir expands the input into many nonlinear, time-mixed features (next slide).

state update: fixed A and W_{in}

$$\mathbf{x}(t+1) = f(\mathbf{A} \mathbf{x}(t) + \mathbf{W}_{in} \mathbf{u}(t))$$

readout: the linear map W_{out}

$$\mathbf{y}(t) = \mathbf{W}_{out} \mathbf{x}(t)$$

training: ridge regression, closed form

X: the states collected in time (one row per time step), Y: the targets, λ : the regularisation.

$$\mathbf{W}_{out} = \mathbf{Y}^T \mathbf{X} (\mathbf{X}^T \mathbf{X} + \lambda \mathbf{I})^{-1}$$

Origins

2001 echo state network (Jaeger)

2002 liquid state machine (Maass, Natschläger, Markram)

2007 “reservoir computing” unifies both (Verstraeten et al.)

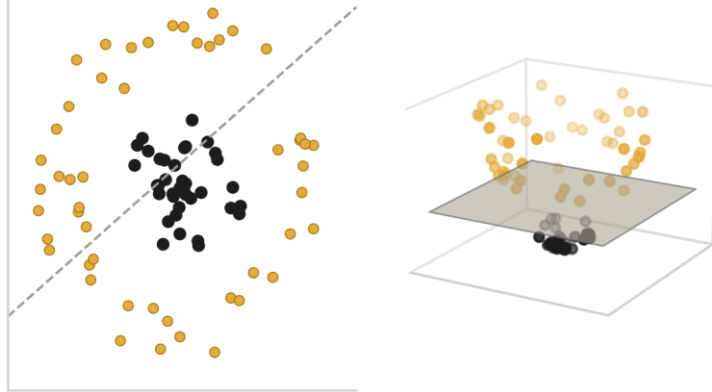
Vocabulary: reservoir = the fixed dynamical system · state = what can be measured of it · readout = the trained linear map.

Only the readout is trained; this is why a physical system can be the reservoir.

1 Projection into high dimension

2-D: no line separates them

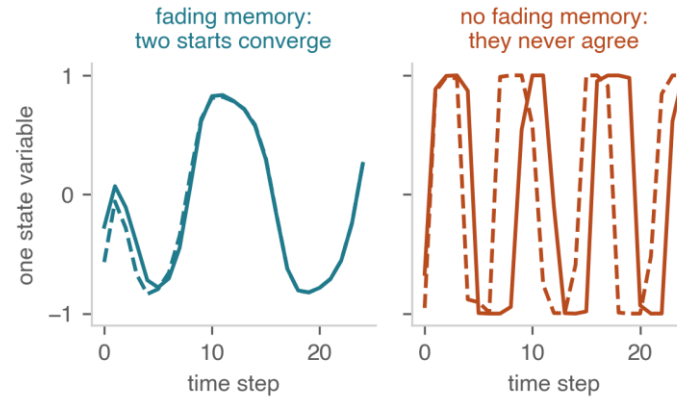
lift $z = x^2 + y^2$: a plane does



- **Two classes**, not separable by a line in 2-D.
- **Lift them nonlinearly** into 3-D: a plane separates them.
- **The reservoir does this** with the input history, so a linear readout is enough.

gives: separability

2 Fading memory (echo state property)

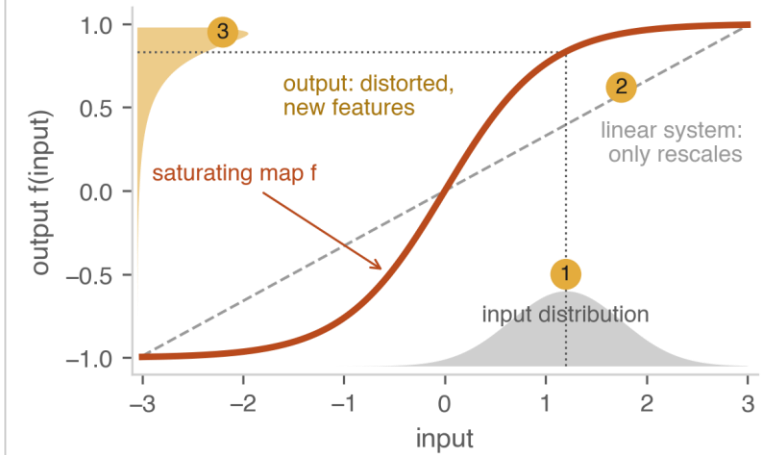


solid and dashed: the same reservoir started from two different states, same input

- **Same input, two initial states:** in a good reservoir the trajectories converge.
- **The state becomes a function** of the recent input, not of the starting point.
- **Without it** the readout fits noise.

gives: a state that depends on the recent past only

3 Nonlinearity creates new features

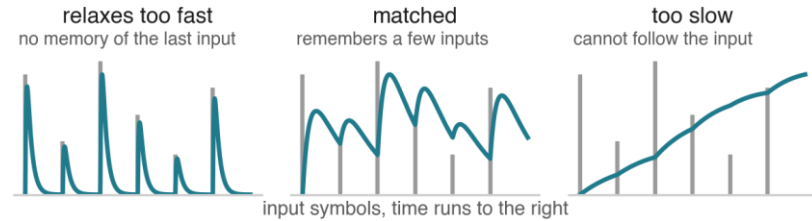


- **Follow** the input distribution is distorted by the saturating map.
- **A linear system** would only rescale it.
- **Saturation creates** harmonics and products of past inputs for the linear readout.

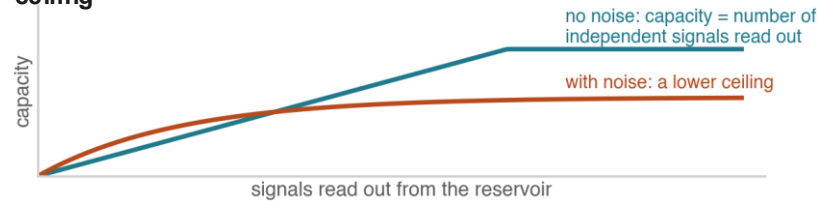
gives: new features, paid for with memory

Everything reservoirs, but not everything reservoirs well

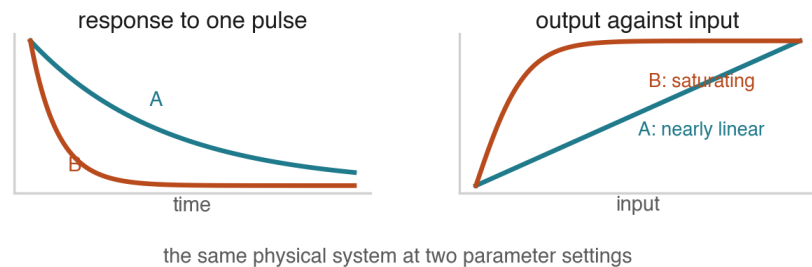
timescales: the system must forget at the rate the input changes



what can be measured bounds the capacity; noise lowers the ceiling



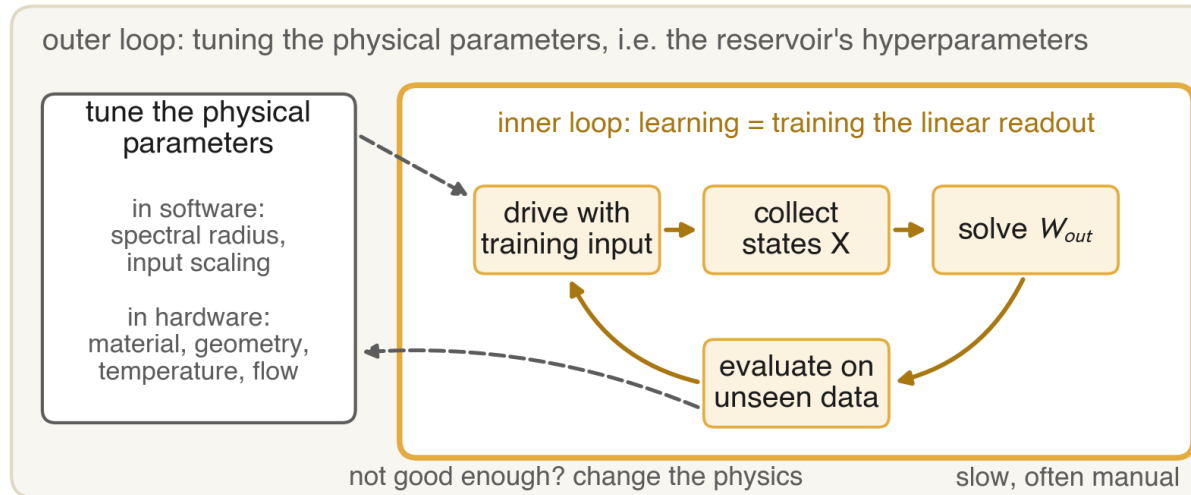
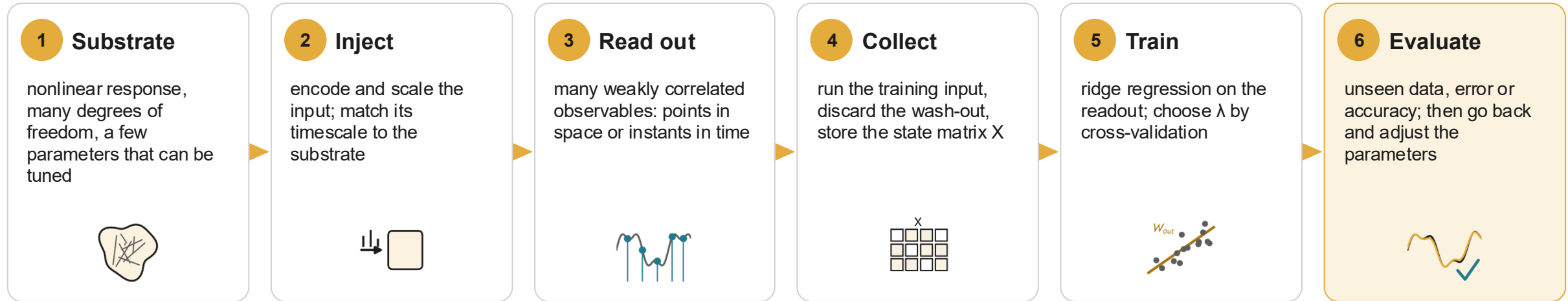
tunability: the same system, different memory and nonlinearity



Six things that decide whether it reservoirs well *Stepney (2024)*

- 1 timescales of input and system of the same order**
it must remember the past for as long as the task needs, and not much longer: too fast, no memory; too slow, it cannot respond to new input
- 2 robustness to environment and internal noise**
the physics is the computer, so its fluctuations are inside the loop; noise caps the usable capacity
- 3 enough observable state**
capacity is bounded by the number of independent signals read out, not by the complexity inside
- 4 a way to excite it**
an input path with enough range to reach the nonlinear regime
- 5 a way to measure it**
a sensor faster than the reservoir's own timescale
- 6 a way to configure or retune it**
a few tunable parameters: the hyperparameters of a physical reservoir

The recipe for a physical reservoir experiment

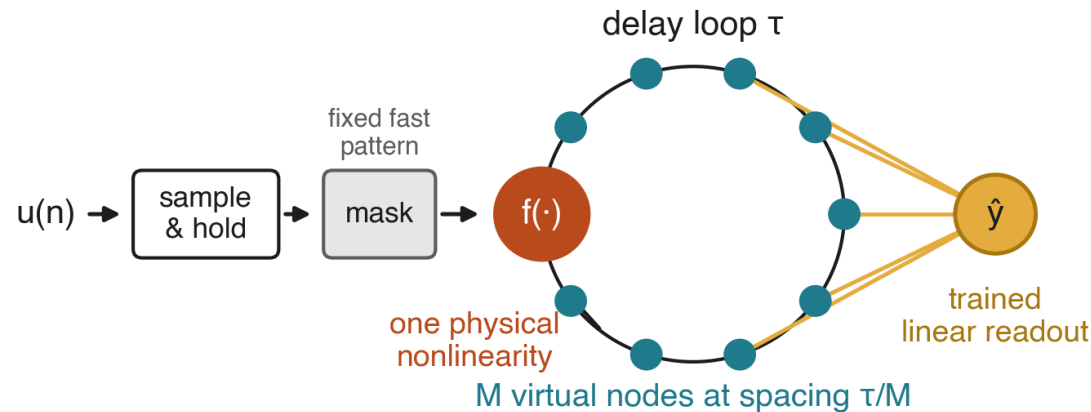


Two loops

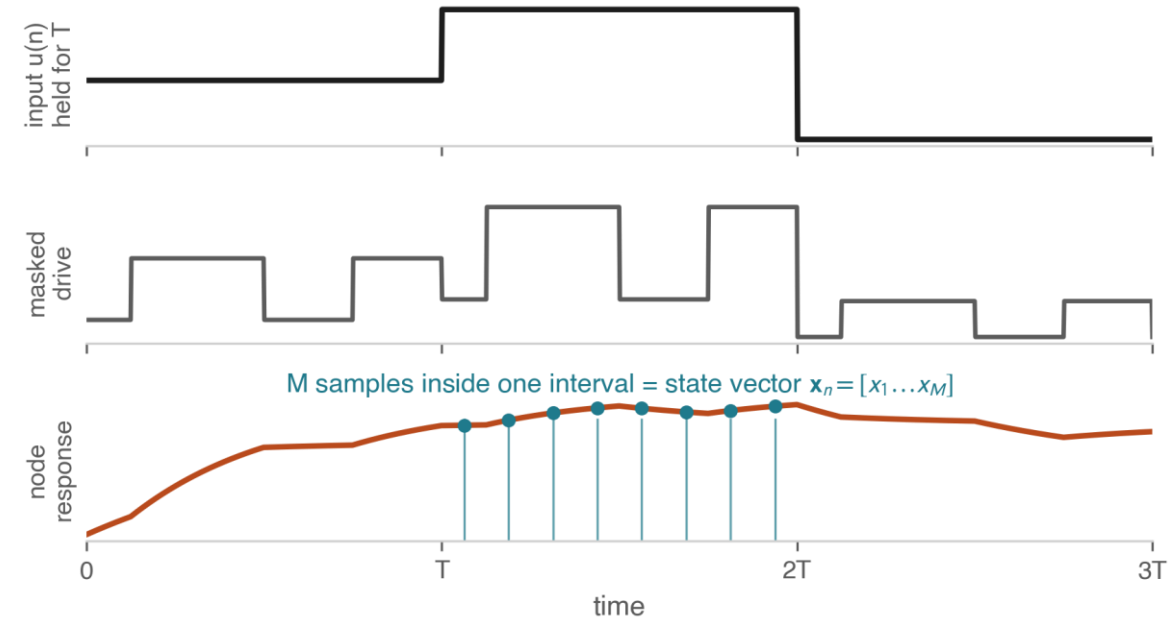
- **Inner loop, learning:** train the linear readout by ridge regression. Fast; it can even be done in chemistry.
- **Outer loop, choosing the operating conditions:** like hyperparameter tuning: slow, often manual; it selects which reservoir we have, it does not train it (spectral radius and input scaling in software; material, geometry, temperature or flow in hardware).
- **In this part:** the parameter maps of our own system are this outer loop.

The two nested loops of a physical reservoir experiment (after Cucchi et al. 2022 and Stepney 2024).

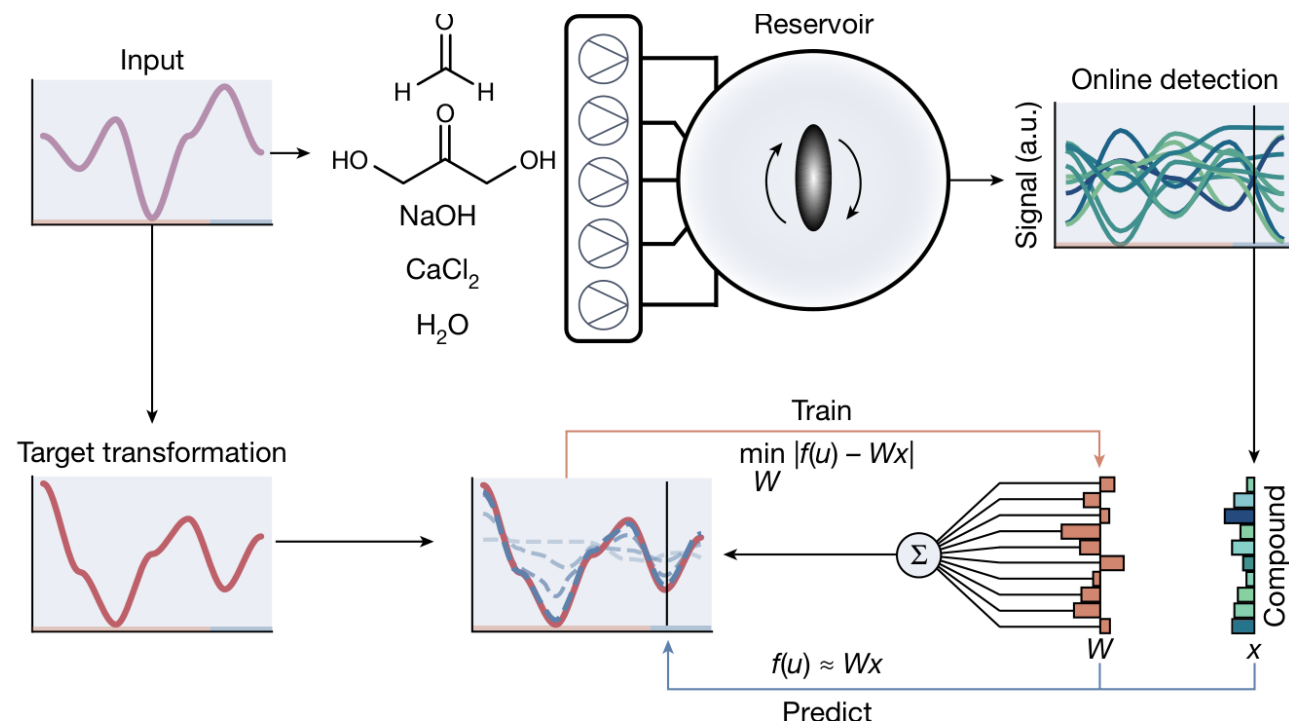
One nonlinear node is enough: time multiplexing and virtual nodes



- **One node plus a delay τ** acts like M virtual nodes spaced τ/M (Appeltant et al. 2011).
- **The input mask**, a fixed random pattern, scales the input differently at each sampling instant; without it the M samples would be M copies of one value.
- **The node's own memory** couples neighbouring samples: it has not relaxed between them.



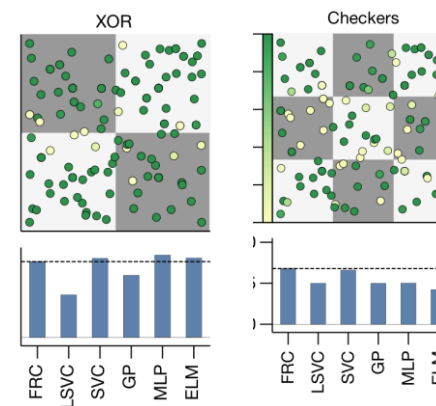
Chemical reservoir computing: a self-organising reaction network



Why the formose reaction? An **autocatalytic** network of hundreds of coupled sugar-forming reactions: rich nonlinear dynamics, memory from the residence time of the reactor, and about a hundred measurable species. These are the properties a reservoir needs, and none of them was designed.

How it is set up *Baltussen et al., Nature 2024*

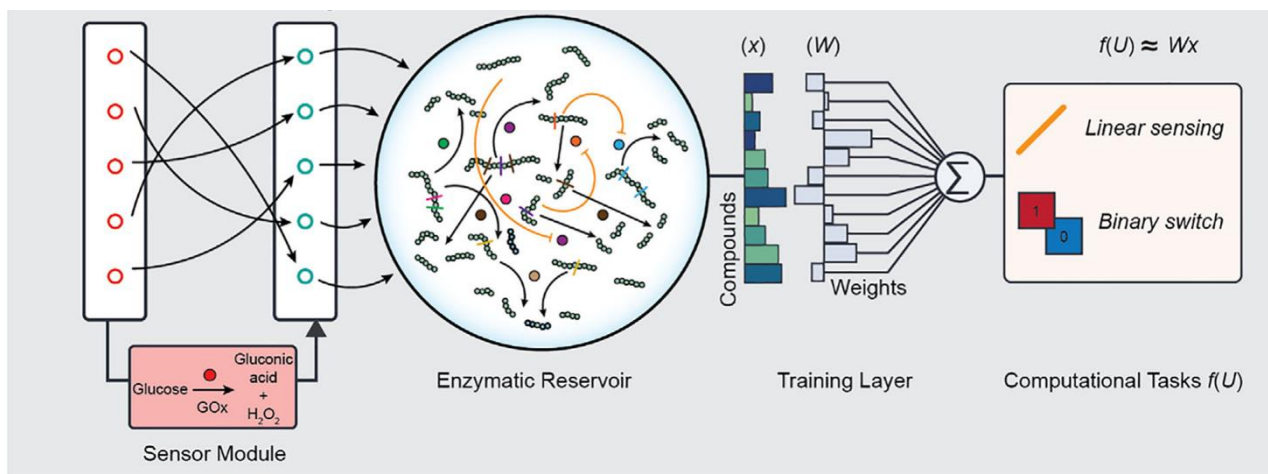
- **Reservoir:** the formose reaction in a stirred-tank reactor.
- **Input:** the flow rates of four pumps.
- **State:** about 100 ion signals from online mass spectrometry, every 0.5 s.
- **Readout:** one linear layer per task, trained by ridge regression.
- **Tasks:** Boolean gates, nonlinear classification, emulation of *E. coli* metabolism, forecasting a chaotic input two minutes ahead.



Nonlinear classification:

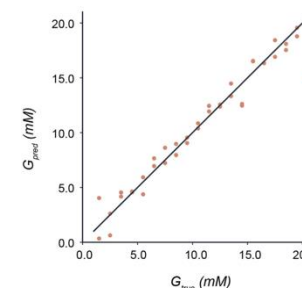
XOR and checkerboard tasks solved from the steady-state composition; the bars compare accuracy with support-vector, Gaussian-process and neural-network classifiers trained on the same inputs.

Reservoir sensing: can a chemical reservoir sense?

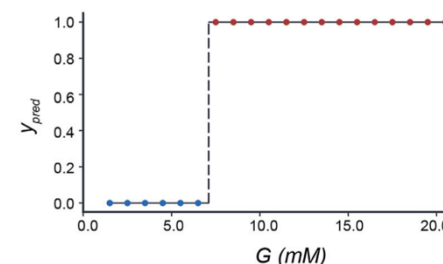


- **Where is the training?** In software: ridge regression or a linear support-vector machine on the measured composition.
- **What the chemistry does:** it senses (glucose $\rightarrow$ pH), stores the recent input in its composition, and expands it nonlinearly.
- **The threshold of the switch** is moved between 4 and 18 mM by retraining only; the chemistry is untouched.

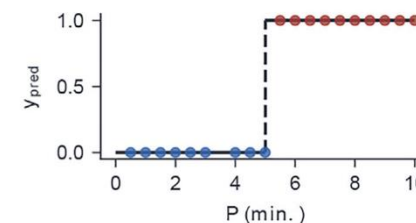
Three tasks from one chemistry



linear glucose sensor
1.5–20 mM, error ± 0.9 mM



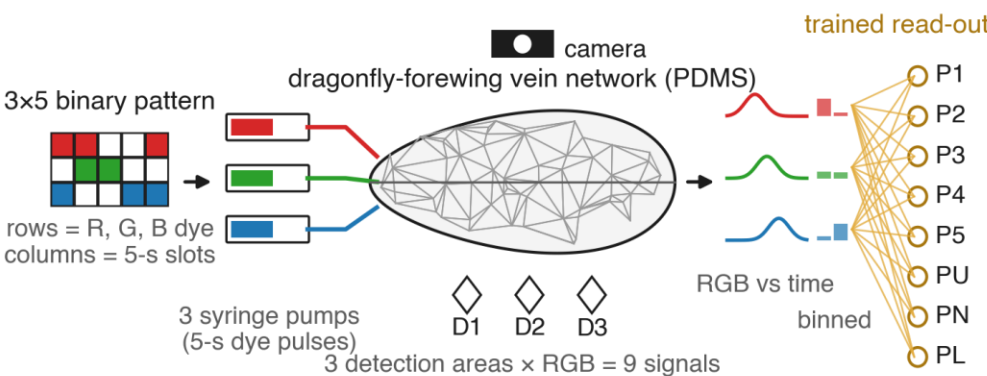
threshold switch at 7 mM
the diabetes threshold;
movable by retraining



temporal pattern
period and amplitude of
glucose pulses

Sensing and computation share one chemical medium.

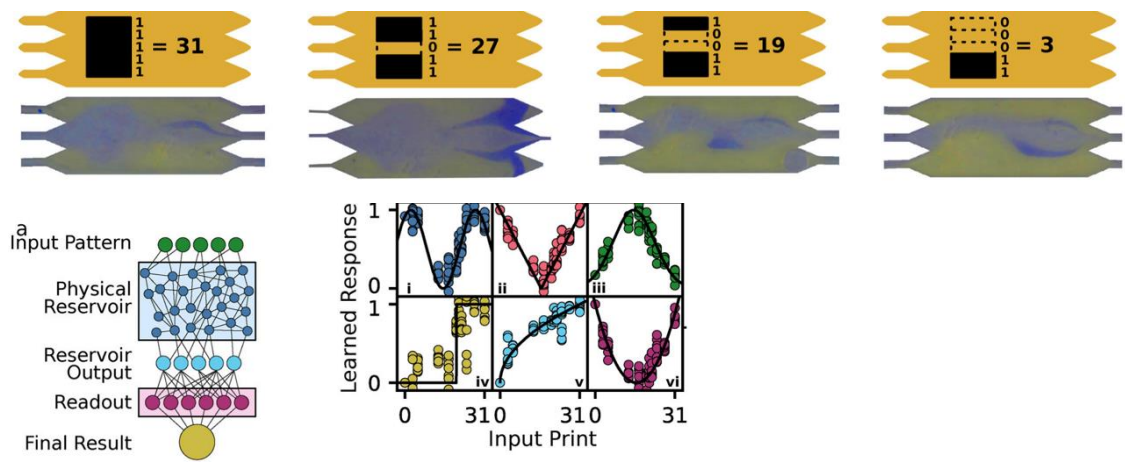
Dragonfly-wing chip *Clouse et al., Device 2026*



- **Reservoir:** the laminar flow in a vein network patterned after a dragonfly wing; dye retained in the veins is the memory.
- **Readout:** nine colour signals from three regions over time feed a small linear classifier: about 90% on eight pattern classes; shuffling the frames in time drops it to 63%.

reservoir laminar flow field	state colour signals in time	tuned by the wing-shaped geometry
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Reconfigurable soft hardware *Swinkels et al., Adv. Mater. 2026*



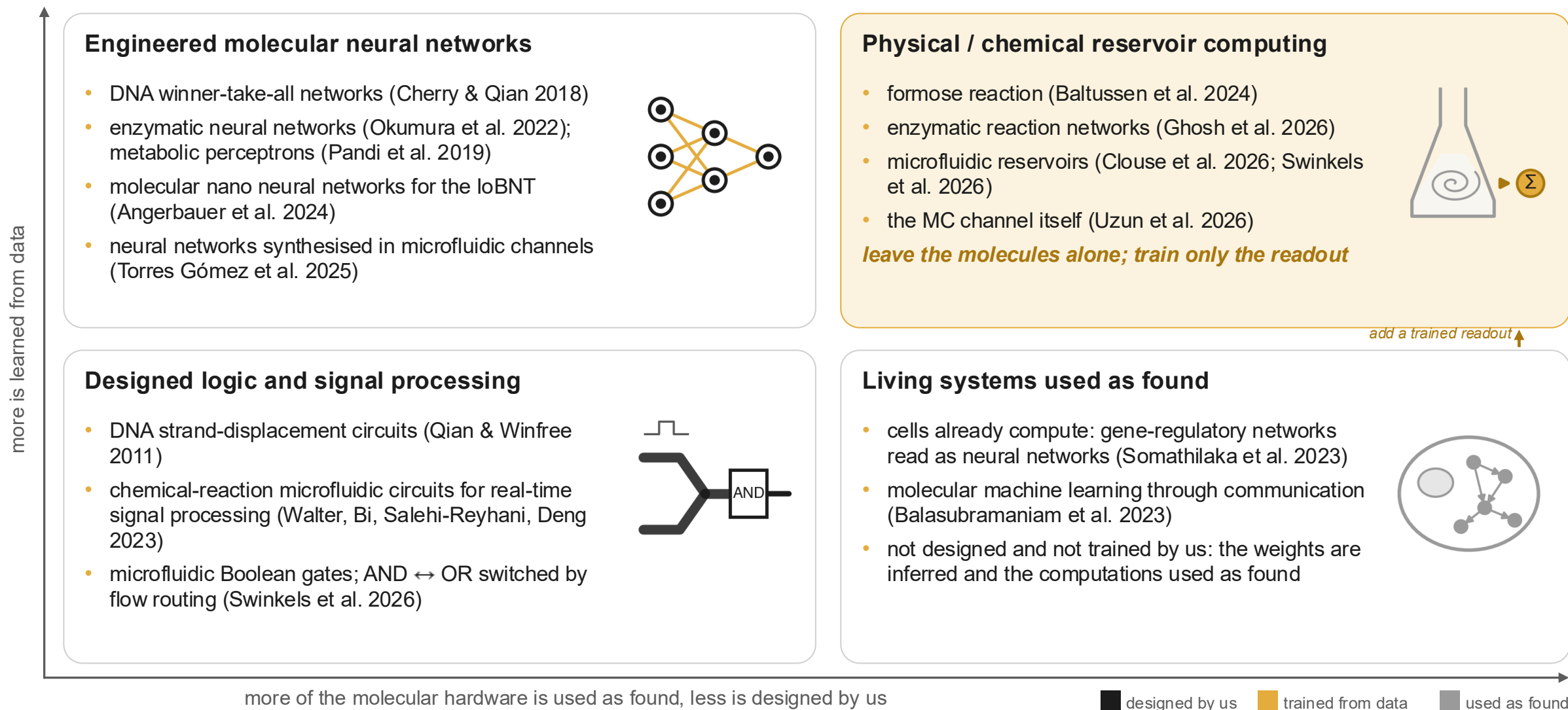
- **Reservoir:** a flowing pH oscillator; its mixing and reaction pattern depend on the geometry printed in the chip.
- **Input and readout:** the printed obstacle pattern is the input, a 5-bit code, printed with light and erased chemically; a linear readout on 20 image features learns nonlinear functions of the code (R^2 up to 0.91).

reservoir mixing pattern	state 20 image features	input printed, erasable obstacles
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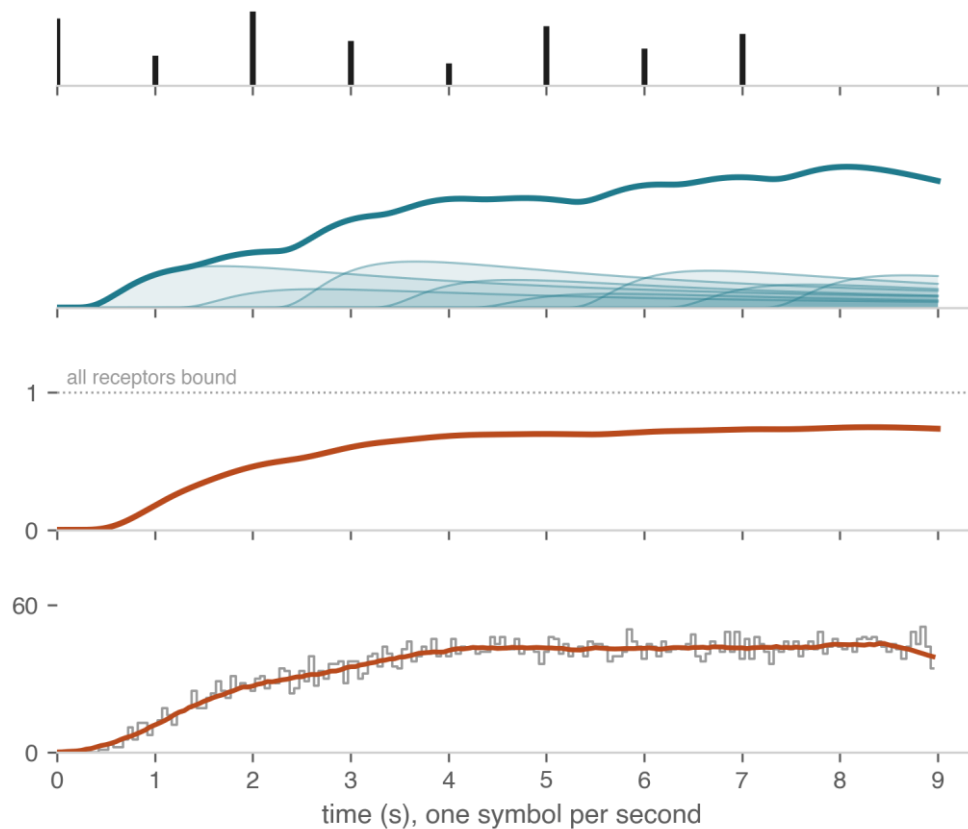
Geometry and mixing set the reservoir's memory and nonlinearity; microfluidics makes both designable.

Left: redrawn from Clouse et al., Device 4, 101027 (2026). Right: adapted from Swinkels et al., Adv. Mater. 38, e73669 (2026).

Where reservoir computing sits among molecular computing paradigms

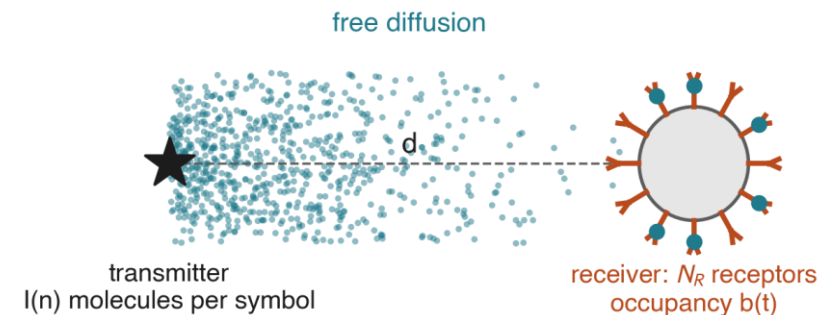


The molecular communication channel: impairments or resources?



Mean-field model of a diffusion channel with a receptor-covered receiver; one symbol per second. Row 4 adds binomial binding noise for 60 receptors.

- 1 released pulses $I(n)$
one per symbol; height = input value
- 2 concentration at the receiver
tails overlap: the receiver sees the past = inter-symbol interference = fading memory
- 3 receptor occupancy $b(t)$
peaks are compressed as receptors fill up = saturation = nonlinearity (plus kinetic memory)
- 4 measured bound receptors
binding is stochastic $\rightarrow$ noise; averaging over about 1 s recovers $b(t)$

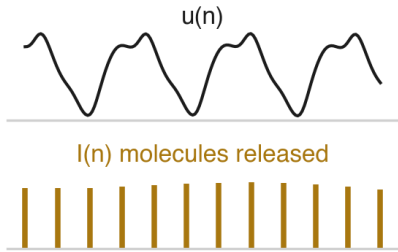


for the MC engineer	for the reservoir	role
inter-symbol interference	fading memory	resource
receptor saturation	nonlinear map	resource
stochastic binding	noise: <i>filter it out</i>	cost

Two problems of the channel are the two things a reservoir needs; the noise remains a cost.

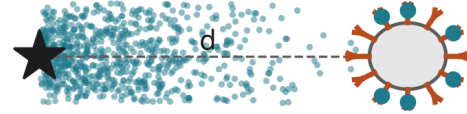
The molecular communication channel as a physical reservoir

1 Input time series



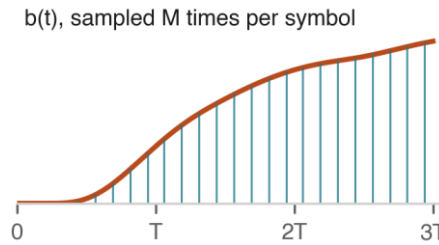
each value $u(n)$ is encoded in the number of molecules released by transmitter in symbol n

2 The channel is the reservoir



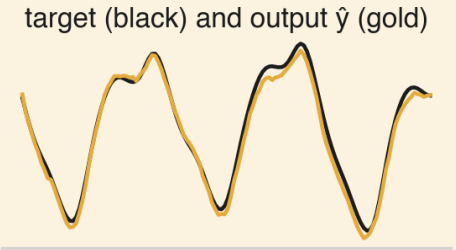
point transmitter · free diffusion · spherical receiver with N_R receptors.
Diffusion = memory,
receptors = nonlinearity.

3 Reservoir state



receptor occupancy $b(t)$, sampled M times inside each symbol: the virtual nodes x_n

4 Linear readout



$\hat{y}_n = W_{\text{out}} x_n$, trained by ridge regression: **the only trained part.**

channel memory: sum of past releases

$$c_R(t) = \sum_n l(n) h(t - nT), \quad h(t) = \frac{\exp(-d^2/4Dt)}{(4\pi Dt)^{3/2}}$$

receptor kinetics: saturation and memory

$$\frac{db}{dt} = k_{\text{on}} c_R(t) [1 - b(t)] - k_{\text{off}} b(t)$$

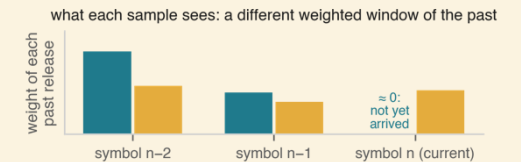
virtual nodes

$$\mathbf{x}_n = [b(nT + \tau_1), \dots, b(nT + \tau_M)]^T$$

readout

$$\hat{y}_n = W_{\text{out}} \mathbf{x}_n$$

No input mask used but why the samples differ: each sampling instant inside a symbol sees a different window of the past, so the virtual nodes are genuinely different.

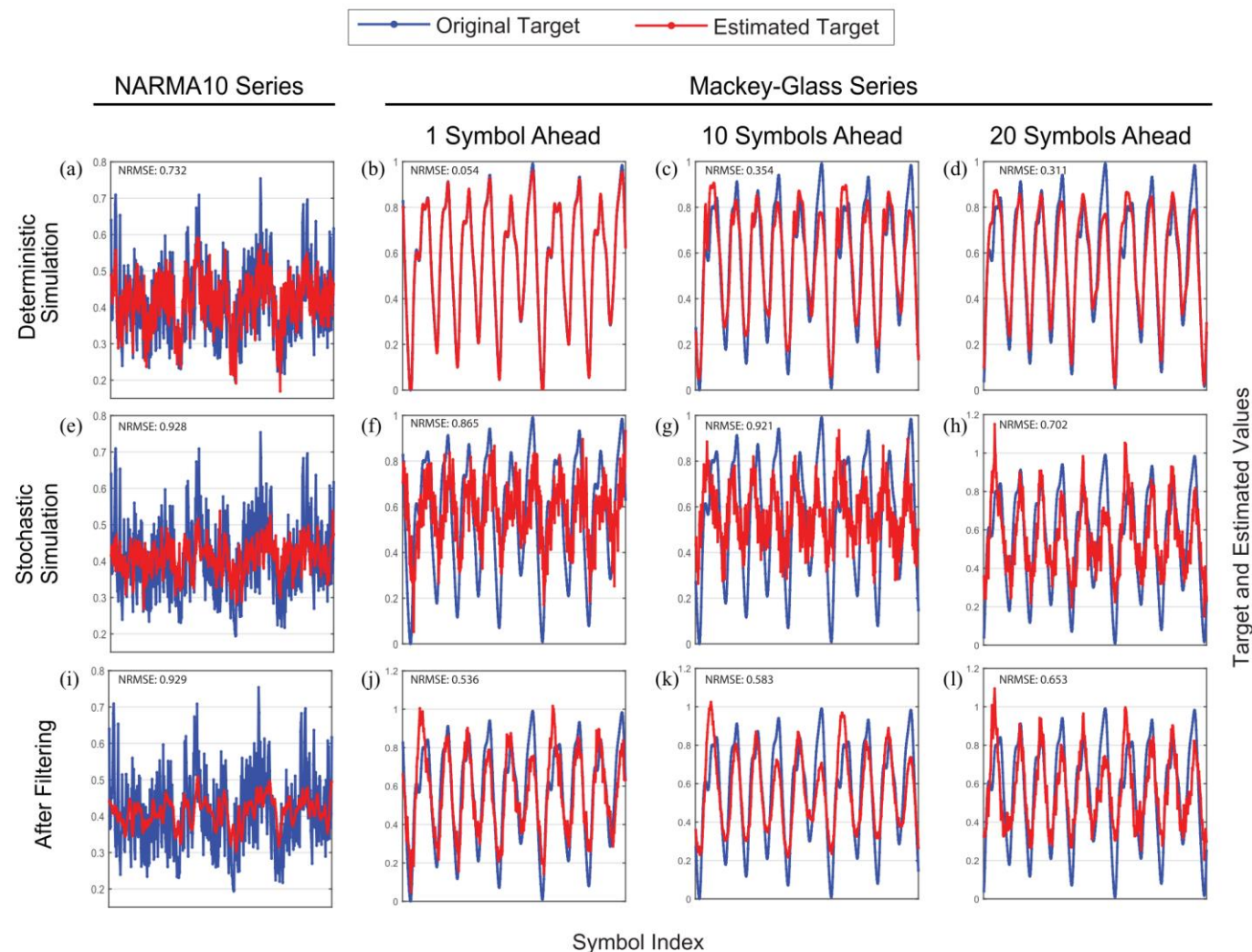


What is different: the reservoir is the communication channel itself; the molecules are the input, the receptor occupancy is the state, and the channel's own physics produces that state.

A simulation study: a mean-field model for the parameter sweeps, and a particle-based model (Smoldyn) with every molecule and receptor for validation.

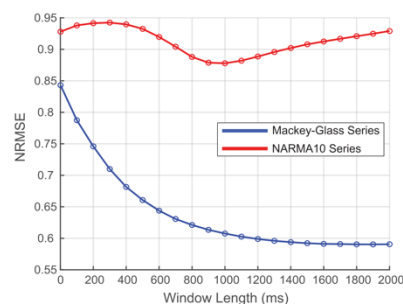
Nothing is engineered: the MC channel's own physics produces the reservoir state.

Does it compute? Benchmarks and the noise floor



error (NRMSE)	Mackey–Glass, 10 steps	NARMA-10	meaning
ideal channel	0.35	0.73	forecasting works; NARMA is weak
with molecular noise	0.92	0.93	noise hides the fine structure
noise + 2 s averaging	0.58	0.93	averaging recovers most of it
software reservoir, same size	0.06	0.40	reference (echo state network)

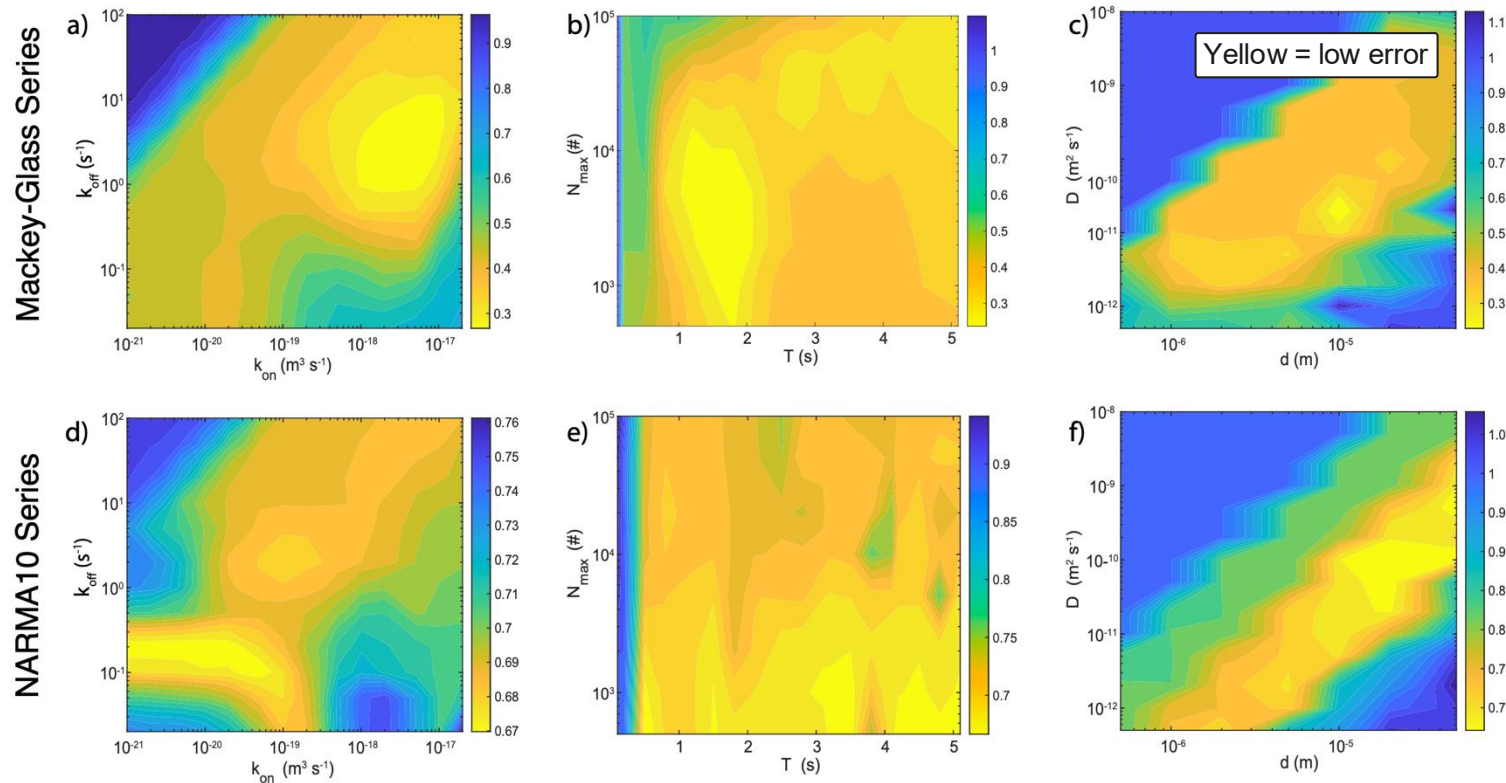
- **Averaging is not a trick added by us:** every real receiver does it. A biosensor has finite bandwidth, and a cell integrates receptor occupancy over time (Berg & Purcell 1977).
- **Why NARMA is weak:** it is driven by white noise and needs fast nonlinear products, which diffusion smooths away. A task–substrate mismatch, not missing capacity.



Averaging window: longer is better for Mackey–Glass; NARMA-10 has an optimum near one second. The window is task-dependent.

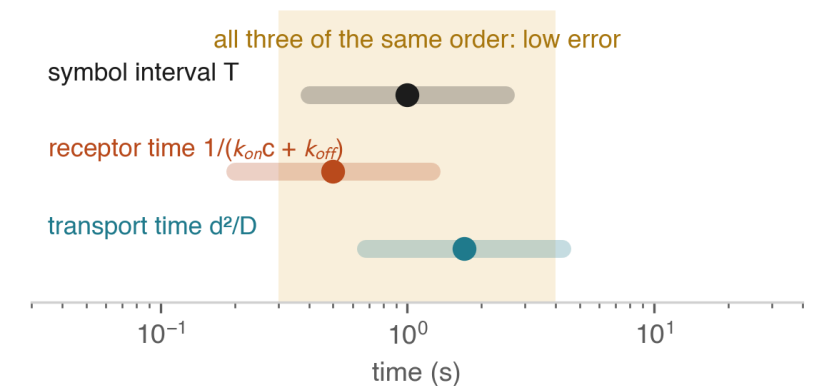
A good reservoir for tasks that match its physics; averaging at the receiver recovers most of what the noise costs.

What governs performance: parameter maps



Match three timescales: transport d^2/D , receptor $1/(k_{\text{on}}c + k_{\text{off}})$, symbol T , all of the same order.

Keep the receptors partly occupied: neither empty (sparse binding) nor saturated (locked).



Default operating point of the paper: the three timescales fall in the same band.

Affinity (k_{on} vs k_{off}): the good region is a diagonal of constant K_D . Too low: sparse binding. Too high: receptors lock.

Pulse vs symbol (N_{max} vs T): below $T \approx 0.5 \text{ s}$ the interference is too strong; longer symbols need stronger pulses.

Transport (D vs d): a diagonal band where d^2/D matches T . Above: too fast, no memory. Below: too weak a pulse.

A task-independent measure: information-processing capacity (IPC)

Benchmarks are task-specific.

A good NARMA score says little about forecasting, and NARMA itself has known problems.

We want to characterise the substrate itself.

How much memory and how much nonlinearity does it offer, and where in parameter space?

IPC (Dambre et al. 2012) does this.

One number per function of the past input, summed into a total that can be compared across systems.

What it is, and how it is computed

Capacity of a target: for a chosen function z of the past input, the fraction of its variance that the best linear readout recovers from the state (0 to 1):

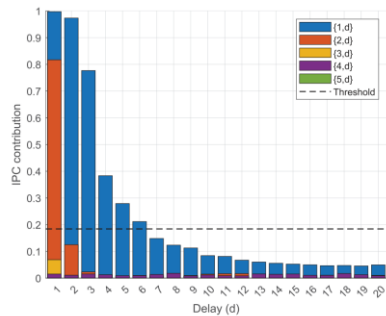
$$C[z] = 1 - \min_{\mathbf{W}} \frac{\langle (z(t) - \mathbf{W} \mathbf{x}(t))^2 \rangle}{\langle z(t)^2 \rangle} \in [0, 1]$$

- 1 **Drive** the reservoir with an i.i.d. random input, uniform in $[-1, 1]$, and record the state $\mathbf{x}(t)$.
- 2 **Targets:** polynomials of delayed inputs, e.g. $u(t-3)$ (linear memory), $u(t-1) \cdot u(t-2)$ (memory and nonlinearity), $P_2(u(t-4))$.
- 3 **Fit** one linear readout per target; drop capacities that are not significant with finite data; sum the rest.

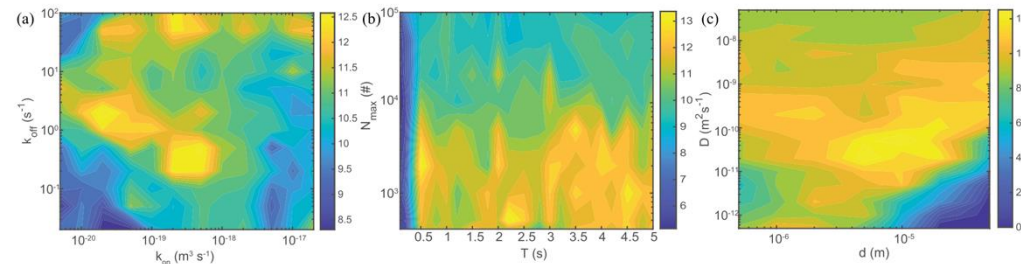
What it tells

- **A fixed budget:** the total capacity can never exceed the number of linearly independent state variables N , and reaches N for any fading-memory system.
- **How the budget is spent:** degree-1 targets measure linear memory, higher degrees measure nonlinear processing.
- **A fingerprint of the substrate:** measured before any task is chosen; noise reduces the usable total.

$$\sum_z C[z] \leq N$$

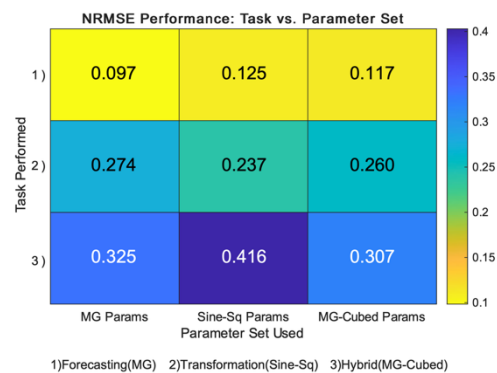
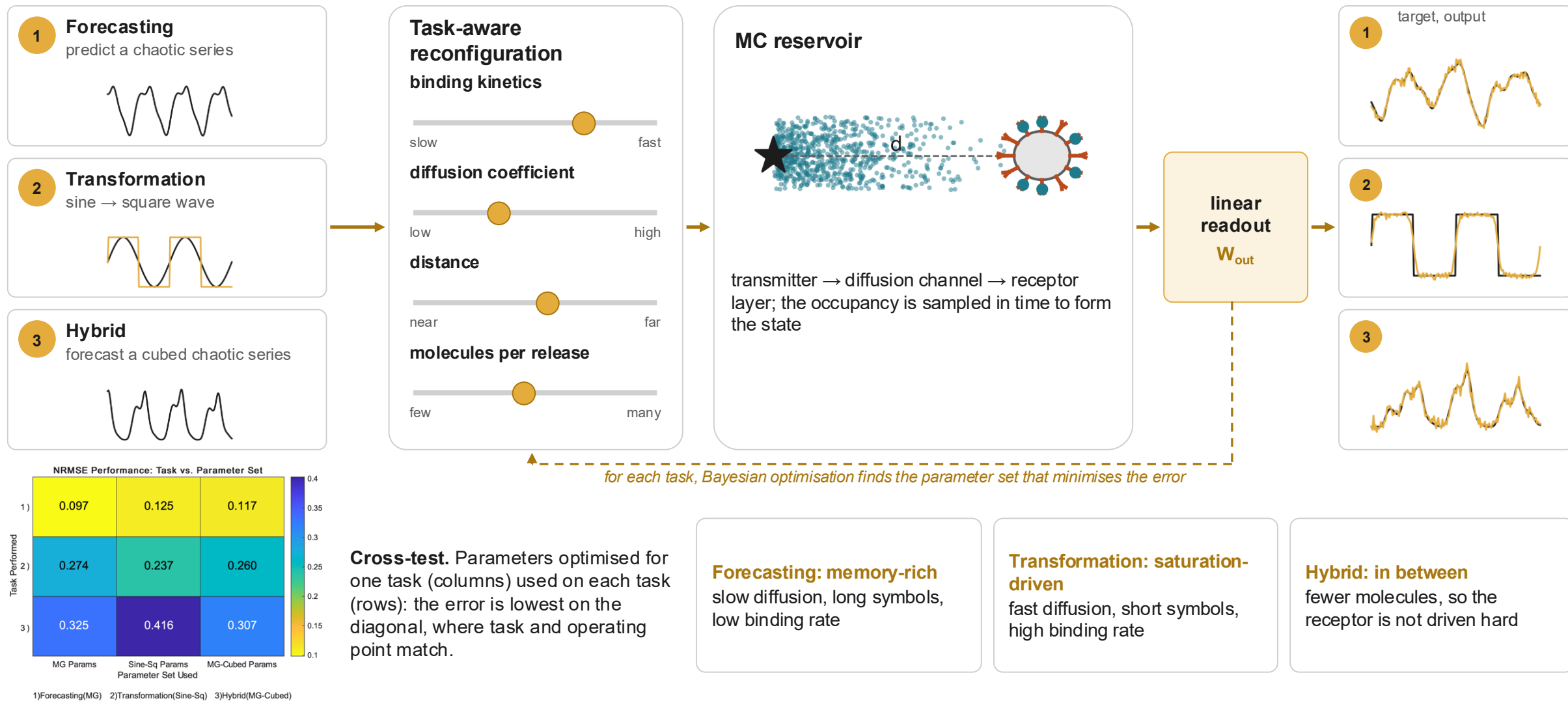


The channel at its default parameters: mostly linear memory of the last three symbols, with second-order terms at short delays; capacity decays with delay.

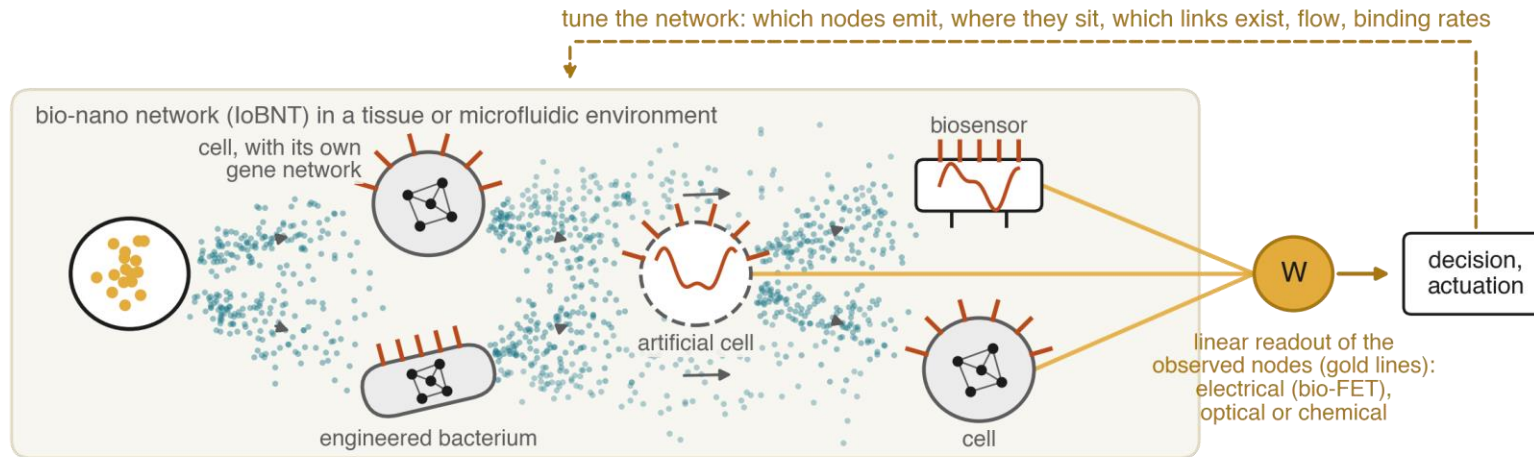


Over the parameter planes: the same shape as the error maps of the previous slide, with the failure regions in the same places.

Task-adaptive MC-PRC: one channel, many tasks



The same channel is a different computer at a different operating point; only physical parameters change.



input node: molecules released here (an emitter, a stimulus, a drug)

the whole network is the reservoir: transmitters, channels and receivers together

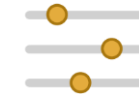
- **The network is a bigger reservoir:** many links between heterogeneous nodes (cells, engineered bacteria, artificial cells, biosensors) give many paths, timescales and nonlinearities, so more signals to read out and more capacity.
- **Biology's own processing comes with it:** cells already process signals through their gene networks and signaling pathways; as nodes of the network they are used as they are.
- **Tuned more flexibly than one link:** which nodes emit, where they sit, which links exist, which nodes are observed. Not every node has to compute: inject at a few nodes, observe a few, and the network in between does the rest.

What is still missing



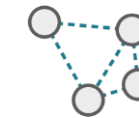
A readout in the wet domain

a bio-FET or an optical sensor now; chemistry has shown sensing (Ghosh et al. 2026) and readout training (Baltussen et al. 2024) in principle.



Tuning the network in place

which nodes emit, where they sit, flow, temperature, competing ligands: the outer loop, inside the body.



Capacity and noise of a network

how memory and nonlinearity add up over heterogeneous nodes, paths and timescales, and how noise adds up over hops, is open.

Design rules, for any chemical or physical reservoir

- 1 **Match the timescales:** the chemistry or the transport must relax on the timescale of the input.
- 2 **Read out enough:** capacity scales with the number of independent signals measured, not with the complexity inside.
- 3 **Drive the nonlinearity, not the saturation:** keep the system in the responsive part of its response.
- 4 **Treat noise as part of the computer:** filter or average at the readout; noise sets the usable capacity.
- 5 **Give the physics tasks it is good at:** smooth, correlated signals for diffusive systems; fast switching for reaction-driven ones.
- 6 **Characterise task-free:** memory and information-processing capacity, not only benchmarks such as NARMA.

Open problems

- 1 **Readout in the chemical domain:** training and computing without electronics, beyond a proof of concept.
- 2 **Rational design:** predicting a substrate's memory and nonlinearity from its physics, and matching it to a task.
- 3 **Reconfiguration in operation:** retuning a running reservoir by temperature, flow, competing ligands or printed geometry.
- 4 **Reliable benchmarks:** NARMA is not comparable across papers; a shared task set for chemical reservoirs is missing.
- 5 **Experimental platforms:** microfluidic testbeds that connect chemistry, flow and readout.

For MC channel: performance is a balance of transport, receptor and symbol timescales at a partly occupied operating point; and one channel can be retuned (even dynamically) for different tasks through its flexible physical parameters.

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Papers Uzun, Ikiz, Kuscu, IEEE Trans. Emerg. Topics Comput. 14(3), 2026 (arXiv:2504.17022) -- Yousuf, Ikiz, Kuscu, arXiv:2602.05931, 2026.

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MC-PRC code

github.com/Kaanikiz/Molecular-Communication-Channel-as-a-Physical-Reservoir-Computer



hands-on tutorial code

github.com/stevenabreu7/handson_reservoir



formose reservoir data

doi.org/10.5281/zenodo.10136537